



Boosting generalization of fine-tuning BERT for fake news detection

Simeng Qin^a, Mingli Zhang^{a,b,c,*}

^a College of Economy and Management, Yanshan University, Qinhuangdao, 066004, Hebei Province, China

^b Hebei Construction Material Vocational and Technical College, Qinhuangdao, 066004, Hebei Province, China

^c Key Laboratory for Health Care with Chinese Medicine of Hebei Province, Shijiazhuang, 066004, Hebei Province, China

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ABSTRACT

Using deep neural networks to detect fake news has gradually become an extremely important field of research. Recent studies, which adopt fine-tuning BERT to detect fake news, have yielded promising results. Unfortunately, fine-tuning BERT is prone to overfitting the training data, which may reduce its generalization ability. The problem remains challenging when one tries to generalize the detector to unseen fake news. First, we investigate the effects of several fine-tuning strategies on the generalization of BERT models used in fake news detection. Specifically, we explore three fine-tuning strategies, which include initialization of model parameters, freezing of model parameters, and training iterations for fine-tuning. Besides, we propose an adversarial fine-tuning strategy to boost the generalization of fine-tuning BERT for fake news detection. In detail, we define the BERT fine-tuning procedure as a mini-max optimization problem. We also propose a novel adversarial fine-tuning training strategy based on feature regularization, called FGM-FRAT, to further improve the generalization. Specifically, we propose to adopt the model representation to guide the generation of adversarial examples during adversarial training on the word vector space. The proposed method outperforms state-of-the-art fake news detection methods. Extensive experiments on several benchmark databases, which include FakeNewsNet, BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, show that using the proposed method improves the generalization accuracy from 61% to 73% for BERT. It indicates that our FGM-FRAT can greatly improve the generalization of fine-tuning BERT for fake news detection. Moreover, the proposed method also can be extended to other pre-trained language models and other text classification tasks.

1. Introduction

Social networking platforms have undoubtedly revolutionized the way individuals access news and information, providing them with a free and user-friendly platform to express their thoughts and connect with like-minded individuals from all corners of the world (Su, 2022). However, in recent years, the downside of this democratization of information dissemination has become increasingly apparent. Fake news has posed a great threat to the public, especially in the 21st century (Howard, Neudert, Prakash, & Vosloo, 2021). Unfortunately, social networking services such as Twitter, and Weibo have contributed to the fake news proliferation. The younger generations, who are still developing their cognitive abilities, are particularly vulnerable to the risks presented by the false information. In particular, the rapid spread of COVID-19 leads to a surge in fake news (Enders, Uscinski, Kloststad, & Stoler, 2020), which will be potentially dangerous for the society (Quinn, Fazel, & Peters, 2021). Therefore, how to effectively detect fake news is becoming increasingly important. And detection of fake news has gradually become an extremely important field of research.

* Corresponding author at: College of Economy and Management, Yanshan University, Qinhuangdao, 066004, Hebei Province, China.

E-mail addresses: qinsm@stumail.ysu.edu.cn (S. Qin), zhangml@ysu.edu.cn (M. Zhang).

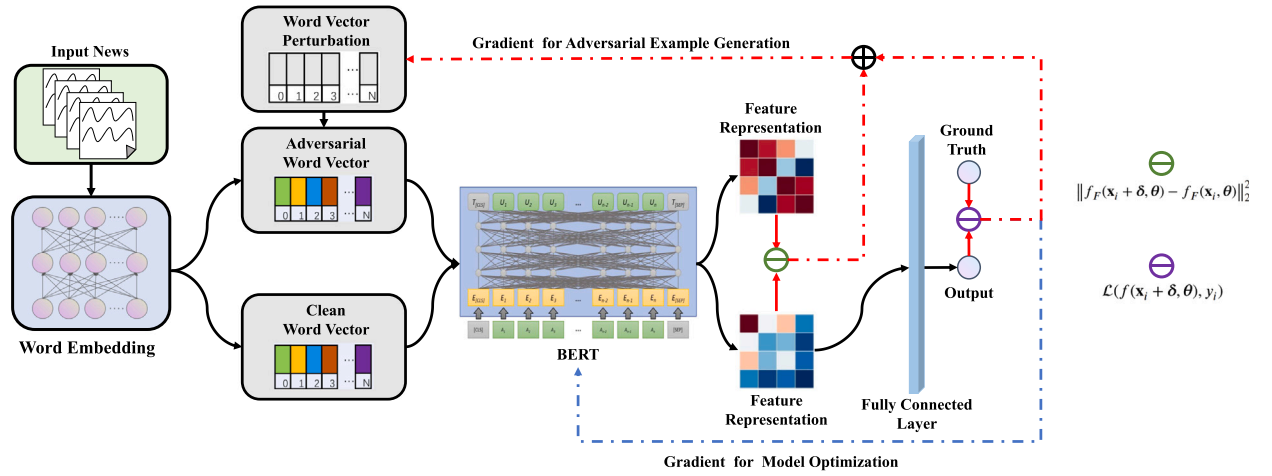


Fig. 1. The pipeline of our adversarial fine-tuning training strategy based on feature regularization, which can be referred to as FRAT. Specifically, we adopt the adversarial regularization loss $\|f_F(x_i + \delta, \theta) - f_F(x_i, \theta)\|_2^2$ and the classification loss $\mathcal{L}(f(x_i + \delta, \theta), y_i)$ to generate adversarial examples. And the classification loss $\mathcal{L}(f(x_i, \theta), y_i)$ is used for model optimization. The adversarial example generation involves using a feature regularization technique to promote the model to learn more general features that are robust to adversarial attacks. By doing so, the model can better generalize to unseen fake news.

Significant progress has been made in developing domain-specific automatic detection of fake news. Especially, deep neural networks have been widely applied in distinguishing fake news (Manzoor, Singla, et al., 2019; Sahoo & Gupta, 2021). Recent works (Serrano, Papakyriakopoulos, & Hegelich, 2020) paid attention to applying transformer-based methods to distinguish fake news, such as Bidirectional Encoder Representations from Transformers (BERT). The emergence of the BERT model has demonstrated remarkable effectiveness in processing and analyzing fake news data. In detail, some works (Kishwar & Zafar, 2023; Varshini, Sree, Srinivas, & Subramanyam, 2023) adopted fine-tuned BERT models to detect fake news and achieve impressive performance in detecting fake news. Besides, some works (Zhang, Wu, Katiyar, Weinberger, & Artzi, 2020) have studied the effects of several fine-tuning strategies, such as initialization of model parameters, freezing of model parameters, and training iterations, on the performance of BERT. Zhang et al. (2020) focused on BERT's contextual representations and identified alternative practices, which solved the instability of the process, improved few-sample fine-tuning with BERT and re-evaluated their effectiveness. Hence, the use of the BERT model is a potential solution to detect fake news.

However, it emerges a significant challenge that the well-trained fake news detection models, including those based on pre-trained BERT, cannot generalize effectively to new and unseen news. In detail, detectors of fake news based on deep learning are only effective on the fake news dataset they were trained on, not unseen fake news. In many cases, these detectors need a significant quantity of training data to learn the distinctive patterns and features that differentiate fake news from real news. However, these detectors may not perform well when faced with new or unseen fake news that have different characteristics or language patterns than the training data. This can lead to inaccurate results and ultimately reduce the overall effectiveness of fake news detector. Recently, many works (Joshi et al., 2023; Wahle et al., 2022) have focused on exploring the generalizability of fake news detector. Particularly, Wahle et al. (2022) systematically evaluate the generalization of BERT-based models on several fake news databases. Through comprehensive experiments, they have indicated that BERT-based models trained on a fake news database underperform on other unseen fake news databases. Thus, while BERT-based models have shown promise in distinguishing fake news, the problem of insufficient generalization remains a significant challenge.

Although previous works have found that BERT-based detectors of fake news suffer from poor generalization, they did not give a solution to improve the generalization of detectors. In this work, different from previous researches, we pay attention to how to improve the generalization of the BERT-based fake news detectors. In detail, we explore the impact of several strategies of fine-tuning on the generalization performance of BERT-based fake news detectors. Specifically, we study the impacts of several strategies of fine-tuning on the generalization of BERT-based detectors used in fake news detection. Our study examines three fine-tuning strategies, including model parameter initialization, freezing of model parameters, and training iterations for fine-tuning. We find that using these fine-tuning strategies for fine-tuning the BERT-based detector can achieve better generalization performance. However, the generalization performance improvement is limited, and it is not enough to identify unseen fake news.

To further improve the generalization of the BERT-based fake news detector, as illustrated in Fig. 1, we propose an advanced adversarial fine-tuning training strategy based on feature regularization aimed at significantly enhancing the generalization of the BERT-based detector used in fake news detection. Specifically, we propose an adversarial training strategy to boost the generalization for the BERT-based detector used in fake news detection. We formulate the process of fine-tuning the BERT-based detector as a mini-max optimization, which can be called adversarial training. The inner maximization optimization generates adversarial examples (Han, Lin, Shen, Wang, & Guan, 2023) for training. The outer minimization optimization boosts the generalization of fine-tuning the BERT-based detector. We find that adversarial training can significantly enhance the generalization of the BERT-based detector used in fake news detection. Actually, the datasets used for fake news detection are typically limited in size. This

characteristic poses a challenge, as fine-tuning BERT on such small datasets may lead to overfitting the source training data. Consequently, this overfitting can result in poor generalization of the BERT model. Hence, to solve the problem, the proposed method adopts FGM to generate adversarial fake news to increase training data diversity for generalization improvement. Building on this insight, we further propose a novel adversarial fine-tuning training strategy based on feature regularization, which can be called FRAT. This strategy is designed to enhance the generalization of the BERT-based detector used in fake news detection. We propose to adopt the model representation to generate the adversarial examples during adversarial training on the word vector space. During fine-tuning, the BERT-based detector of fake news is trained to recognize and correctly classify these adversarial examples, which improves its ability to handle similar examples in the future. By incorporating feature regularization into the adversarial fine-tuning training strategy, the model is able to better generalize to new and unseen data, leading to improved generalization performance for fake news detection. The proposed method can outperform state-of-the-art fake news detection methods. Besides, the proposed method can be extended to other pre-trained language models such as RoBERTa to improve the generalization performance for fake news detection. The proposed method also can be applied to other text classification tasks such as movie review sentiment analysis, which enhances the potential impact of the proposed method.

In summary, our research offers a thorough examination of how various fine-tuning strategies can influence BERT's generalization in detecting unseen fake news. By exploring both conventional and adversarial fine-tuning approaches, we can significantly improve the BERT generalization in detecting fake news. By incorporating feature regularization into the adversarial training process, we can further improve the generalization of fine-tuning BERT. Our proposed strategy has the potential to significantly enhance the generalization of the fine-tuning BERT-based detector of fake news, ultimately contributing to the development of more accurate and reliable models that can effectively combat the spread of misinformation. Moreover, the proposed method also can be applied to other text classification tasks to improve the model generalization such as sentiment analysis, which enhances the versatility of our method.

Our study has made significant contributions in four key aspects:

- Firstly, we have conducted a thorough analysis of the effects of several fine-tuning strategies on the generalization of the fine-tuning BERT-based detector of fake news. Specifically, we explore the effects of initialization of model parameters, freezing of model parameters, and training iterations for fine-tuning and find using these fine-tuning strategies only can slightly improve generalization for fake news detection.
- Secondly, we propose an adversarial training strategy for the fine-tuning BERT-based detector of fake news. From the experimental results, we discover that the implementation of adversarial training could boost the generalization of the BERT-based detector of fake news.
- Thirdly, we propose a novel adversarial fine-tuning training strategy based on feature regularization to further improve the generalization of fine-tuning BERT, called FGM-FRAT. The proposed method has been shown to effectively enhance the generalization performance of the BERT-based detector of fake news. Moreover, our method can be used for other pre-trained language models and applied to other text classification tasks.
- Finally, extensive experiments are conducted on several benchmark databases, which include FakeNewsNet, BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews. The results can improve the accuracy from 61% to 73% for BERT. And the proposed method outperforms state-of-the-art fake news detection methods. The result demonstrates the superiority of our FGM-FRAT. Besides, the proposed method can also be used to improve generalization for other text classification tasks.

2. Related work

We first present the related works about deep learning techniques for fake news detection. Then, researches on fine-tuning BERT-based fake news detectors are introduced. Next, we introduce the related researches on domain generalization-based fake news detection methods. Finally, we present the related works about generalizability methods on text classification tasks.

2.1. Deep learning methods

Fake news detection has garnered significant attention most recently, leading many scholars to employ deep learning techniques to attain improved precision (Shone, Ngoc, Phai, & Shi, 2018). In particular, many recent studies have focused on utilizing microblogs and comment features to detect fake news. Through extracting the language patterns, sentiment, and other characteristics of social media posts and comments, researchers have been able to develop deep learning models that can effectively detect fake news (Zhang & Ghorbani, 2020). These models adopt neural networks and deep learning algorithms that could learn to recognize the patterns and characteristics that distinguish fake news, and differentiate them from legitimate news sources. The use of microblogs and comment features in fake news detection has proven to be particularly effective (Bondielli & Marcelloni, 2019). By analyzing the content and context of social media posts and comments, researchers are able to detect fake news. Specifically, several works (Chen, Li, Yin, & Zhang, 2018) have focused on utilizing convolutional neural networks and recurrent neural networks to learn hidden representations from text, enabling the capture of variations in text and user representations over time. Shu, Cui, Wang, Lee, and Liu (2019) proposed to adopt hierarchical attention networks (HANs) to detect fake news. By introducing a hierarchical attention mechanism, HANs can effectively record the semantic relationships between different parts of the text, thereby achieving better detection performance. Ma, Gao, and Wong (2018) have proposed a novel approach that incorporates structural awareness into recursive neural networks to achieve more accurate and effective detection of fake news. By taking into account the structural

features, this method achieves a more robust and reliable classification of fake news. Other studies have also explored more innovative and cutting-edge methods to detect fake news, such as adopting multi-head attention (Yuan, Ma, Zhou, Han, & Hu, 2020) or graph neural networks (Bian et al., 2020). These approaches all share the same objective: to leverage the unique properties of various deep neural network architectures to achieve better performance in detecting fake news. Specifically, these methods aim to exploit the powerful learning and representational capabilities of these models to capture the nuanced and complex features inherent in fake news and to distinguish it from legitimate news. Additionally, some works adopted pre-trained language models to detect fake news. Specifically, Praseed, Rodrigues, and Thilagam (2023) proposed to adopt pre-trained language models, such as BERT, to achieve advanced performance to detect fake news. The main idea behind this approach is to fine-tune a pre-trained language model using a large corpus of news articles, with the aim of enhancing the model's ability to accurately differentiate between fake news and true news. Through fine-tuning the language models, the model can learn highly sophisticated representations of language, and capture the complex and subtle patterns of fake news. Overall, detecting fake news using deep learning methods has demonstrated great promise, with a wide range of methods being explored to achieve the effective results.

2.2. Fine-tuning of BERT

Deep Neural Networks (DNNs) are widely employed in a variety of applications due to their excellent performance in handling complex tasks (Zheng, Yahya, Wang, Zhang, & Hoshyar, 2023). However, one of the fundamental challenges in using DNNs is that they need lots of training data to complete learning. This is particularly true for tasks that involve language understanding, where the magnitude of the dataset can have a substantial effect on the model's effectiveness. Fake news detection is no exception to this challenge. Currently, available fake news datasets are limited in size, with most containing only a few thousand samples. Small datasets are far from enough to train deep neural networks, and this is where transfer learning can come into play. Transfer learning is a technique that uses previously trained models for downstream tasks, thus decreasing the quantity of data necessary for training. The BERT is widely applied in detecting fake news (Aljawarneh & Swedat, 2022). By using the expertise learned from extensive training data, pre-trained models can obtain advanced performance on small fake news datasets. The prevailing method of utilizing the pre-trained BERT model is to replace the initial output layer with a new layer that is customized for the particular task, and then refining the entire model, *i.e.* fine-tuning of BERT. This requires knowledge of the parameters of the new output layer. All weights of the model are altering, including those of the pooler, transformer blocks, and word embeddings. Previous works (Sultana & Nishino, 2023), which adopt BERT to detect fake news, have shown that using BERT can significantly enhance the effectiveness of fake news detection. However, these works have studied the specific application of BERT for detecting fake news but have not explored the generalization.

2.3. Domain generalization-based fake news detection methods

Fake news arises across diverse domains of the Internet. Detecting fake news in multiple domains can be called Multi-domain Fake News Detection (MFND). Existing Multi-domain Fake News Detection mainly has two main problems: 1. Heterogeneous Training Data: incorrect messages in the training set come from different areas, which could lead to a decrease in the model's performance. This phenomenon may be attributed to the distinct content and style inherent in each domain (Nordberg, Kävrestad, & Nohlberg, 2020). 2. Insufficient Data in Single Fields: within a singular domain, the availability of limited data poses a challenge to training a robust fake news detection model. Some works adopt adaptation-based or domain generalization-based methods to solve the above problems for fake news detection. For example, some researchers (Liang, Zhang, Li, Zhang, & Yu, 2022; Nan, Cao, Zhu, Wang, & Li, 2021) have indicated that it is hard to detect the fake news from different domains due to differences in the meaning of professional words, the unique sentence structures and contents in different domains. Nan et al. (2021) proposed to enhance the input by appending suitable single-domain labels to each news to improve detection performance, which can be called multi-domain fake news detection model (MDFEND). In detail, MDFEND extracts the common characteristics of fake news in all different domains and adopts the single domain label to detect the specific characteristics of fake news from a certain domain. Liang et al. (2022) proposed to adopt a fuzzy inference method to provide a series of labels for news, which can be called Fuzzy-domain Fake News Detection Model (FuDFEND). FuDFEND employs a neural network for fuzzy inference, generating fuzzy domain labels for news items. The feature extraction module then utilizes these labels to extract multi-domain features and generate a total feature representation. MDFEND and FuDFEND achieve significant improvements in multi-domain fake news detection tasks. Although they explore how to effectively deal with fake news in different domains, the above studies are still limited to the same dataset and do not explore model generalization across different datasets.

2.4. Generalizability methods on text classification tasks

A series of works (Buyukoz, 2023; Qi, Chen, Mao, Jia, Duan, Zhang, & Xue, 2023; Zanzwar, Wiechmann, Qiao, & Kerz, 2022) have paid much attention to the generalizability on text classification tasks such as fake news detection, natural language understanding, text emotion detection, sentiment analysis, etc.

In the domain of fake news detection, more works (Ni, Guo, Li, & Jiang, 2020; Pennycook & Rand, 2021; Suprem, Vaidya, & Pu, 2022) focused on studying the generalizability of deep learning models for fake news detection. Suprem et al. (2022) studied the generalization of some transformer-based fake news models and find that the fine-tuned models cannot generalize well to unseen news, which does not overlap the training data. And then they propose to adopt a KMeans-Proxy to boost generalization performance.

KMeans-Proxy divides the embedding space into some local regions and leverages local characteristics to establish a rejection mechanism for fake news detection models. This approach enhances generalization performance by effectively incorporating localized information. [Stepanova and Ross \(2023\)](#) focused on COVID-19 fake news datasets and propose a supervised content-based, which adopts Generative Adversarial Network (GAN) architecture to craft conditionally plausible representations of fabricated news, to boost the model's generalization for fake news detection.

As for the domain of natural language, [Chen, Shen, Chen, and Yang \(2021\)](#) have proposed to adopt an effective data augmentation method to enhance the model generalizability for the natural language understanding tasks. [Kim, Park, and Han \(2022\)](#) studied the generalization of implicit Hate Speech Detection and proposed to use contrastive learning to improve performance. Specifically, they proposed an implication-based contrastive learning method (ImpCon) by pulling an implication and its corresponding hateful posts. Experimental results indicate that fine-tuning with ImpCon could enhance generalization ability.

Regarding the domain of text emotion detection, [Zanwar et al. \(2022\)](#) explored how to effectively improve the generalization ability of text emotion detection. In detail, they proposed to adopt transformer models, including BERT and RoBERTa, which are enriched with psycholinguistic features to improve the generalizability of text-based emotion detection. In an effort to bridge the research gap in Arabic text emotion detection, [Abdaljalil, Hassanein, Mubarak, and Abdelali \(2023\)](#) conducted a comprehensive study on the generalization of machine learning models using a dataset of approximately 1.64M Arabic tweets, employing various techniques, such as Support Vector Machines (SVM), Convolutional Neural Networks (CNNs), and Long Short-term Memory (LSTM), etc, to enhance model performance across unseen data.

When it comes to the domain of sentiment analysis, [Seng and Wu \(2023\)](#) focused on the generalizability of the sentiment analysis tasks. Specifically, they developed a dual-stream network designed to strategically redirect attention from words that carry contrasting sentiments, effectively avoiding biased judgments. The proposed method can effectively improve the generalization performance of sentiment analysis models. [Abdaljalil et al. \(2023\)](#) conducted a comprehensive study on model generalization for Arabic sentiment analysis and focused on studying the techniques to boost model generalization. They have demonstrated that models trained on data refined through a cascaded approach have achieved the best generalization performance. Hence, the generalizability of text classification tasks becomes more and more important, ensuring they can adapt to and perform well on a wide array of datasets and linguistic contexts without the need for extensive retraining. This focus on generalization not only improves the efficiency and effectiveness of model deployment in real-world applications but also contributes to the advancement of natural language processing technologies as a whole. In this paper, we focus on boosting the generalization of fine-tuning BERT for fake news detection.

3. Research objectives

The absence of labels in the fake news dataset caused its limited scale, leading to the suboptimal generalization performance for traditional deep learning methods based on supervised learning. The pre-trained BERT models exhibit a significant advantage in improving generalization on such small datasets as the fine-tuning of BERT introduces additional knowledge to help extract more abundant content features. However, it still has gaps. The essential problem of the bad generalization is that the lack of labels leads to the lack of supervised training, while the pre-trained BERT does not solve it. To outcome the issue, we propose an adversarial training strategy based on FGM, called FGM-FRAT, to increase the amount of training data with labels, which enables the fine-tuning of BERT to comprehensively extract content features, thereby further improving the generalization of BERT for detecting fake news. Therefore, the main objectives of the proposed FGM-FRAT are as follows: (1) To explore the impacts of several fine-tuning strategies on the generalization of the BERT-based fake news detectors. (2) To study a carefully designed adversarial training strategy to generate labeled training data to obtain richer semantic features for fake news detection. (3) To leverage feature regularization to enhance the generalization performance of the BERT-based detector of fake news. (4) To evaluate the effectiveness of the proposed method for improving the generalization of fine-tuning BERT to detect fake news across diverse datasets.

4. Research motivation and novelty

Research motivation: This work is motivated by the critical challenge that existing fake news detection models, especially those based on the pre-trained BERT (Bidirectional Encoder Representations from Transformers) framework, struggle with generalizing to new and unseen fake news. Despite the proven effectiveness of BERT-based models in detecting fake news within the datasets they were trained on, these models often fail to maintain their performance when encountering fake news that exhibit different characteristics than those in the training data. This limitation not only undermines the accuracy of fake news detection but also diminishes the overall reliability of such detectors in real-world scenarios where the nature of fake news is constantly evolving.

Research novelty: Our research introduces novel contributions in the field of fake news detection with BERT-based models, focusing on improving their generalization to unseen data: (1) Comprehensive Analysis of Fine-tuning Strategies: We conduct a detailed analysis of how different fine-tuning strategies, including model parameter initialization, freezing, and training iterations, slightly improve the generalization of BERT-based fake news detectors. (2) Adversarial Training Strategy: We innovate by proposing an adversarial training strategy that significantly enhances the model's generalization capability, showing a clear improvement in detecting unseen fake news. (3) Feature Regularization-based Adversarial Fine-tuning: Our novel Feature Regularization-based Adversarial Fine-tuning (FGM-FRAT) strategy further boosts generalization, demonstrating effectiveness not only for fake news detection but also for other text classification tasks. (4) Extensive Validation Across Multiple Datasets: Through experiments on multiple datasets, we prove our methods' superiority by improving detection generalization accuracy significantly and outperforming current state-of-the-art approaches. Together, these contributions represent substantial advancements in the field, offering novel methodologies and insights that pave the way for more robust, adaptable, and effective fake news detection solutions.

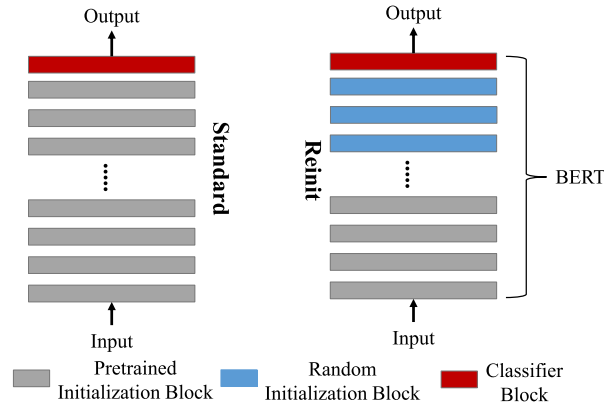


Fig. 2. The Reinit strategy aims to explore the effect of initializing with the last few layers of the pre-trained BERT during the fine-tuning. All the gray layers gather together constructing to BERT, and each gray layer represents a bidirectional Transformer block within BERT.

5. Method

First, we explore the effects of fine-tuning strategies on the generalization of the BERT to detect fake news. They include initialization of the BERT's layers in Section 5.1.1, freezing of the BERT's layers in Section 5.1.2, and training iterations for fine-tuning in Section 5.1.3. Then, by assembling these fine-tuning strategies, we conclude a baseline for BERT to detect fake news in Section 5.1.4. Moreover, we propose an adversarial fine-tuning training strategy in Section 5.2.

5.1. Impact of fine-tuning strategies on the generalization

Fine-tuning is a wide machine-learning technique that can improve the efficiency of pre-trained models on specific tasks. In particular, when applying fine-tuning strategies, the pre-trained BERT is adjusted on the fake new detection dataset to improve performance. However, while fine-tuning can lead to impressive results on a specific dataset, it may not generalize well to new and unseen news. This is because parameters of the model are updated during the fine-tuning process on a particular dataset and task, leading to overfitting and limiting the model's ability to generalize well to unseen data. We aim to address the issue of fine-tuned BERT's poor generalization performance by exploring the impact of three different fine-tuning strategies: initialization, freezing, and training iterations. Specifically, the pre-trained BERT is fine-tuned on the FakeNewsNet dataset, which contains 44,000 records of both fake and real news data. To test how well the trained models can generalize on an unseen dataset, we use the BuzzFeedNews dataset, which includes 2,282 records of fake and real news. Section 6.1 provides a thorough description of these datasets.

5.1.1. Initialization of the pre-trained BERT layers

The importance of the initial values of model parameters cannot be overstated. These values determine the starting point for the optimization process, and thus have profound effects on the final model performance. Several studies (Dauphin & Schoenholz, 2019) have demonstrated the crucial role of initialization methods in achieving good performance. In the context of fine-tuning pre-trained models like BERT, the selection of the initialization approach can greatly influence the ultimate efficacy of the model during fine-tuning. The fine-tuning model can leverage the knowledge learned during pre-training and adapt it to the current task to improve generalization. As illustrated in Fig. 2, in the standard method of fine-tuning BERT, all layers are initialized with pre-trained weights of BERT except for one specialized output layer. Some research (Zheng et al., 2022) has shown that the lower layers of pre-trained models typically acquire more generalized characteristics, while the upper layers located closer to the output become more specialized in the pre-training tasks. However, despite the widespread use of pre-trained models in detecting fake news, the influence of their layer initialization on the generalization of these models has not been extensively investigated. Therefore, in order to shed light on this important issue, we undertook an extensive investigation to determine how pre-trained BERT initialization affects the performance of fake news detection tasks. By conducting a rigorous ablation study, we aimed to determine the specific effects of different initialization techniques on the overall generalization capabilities of BERT-based models in this domain. Note that in this paper, we focus on the initialization of task-specific models, i.e., BERT-based fake news detection models instead of the initialization of parameters of pre-trained BERT models. The Initialization is shown in Fig. 2.

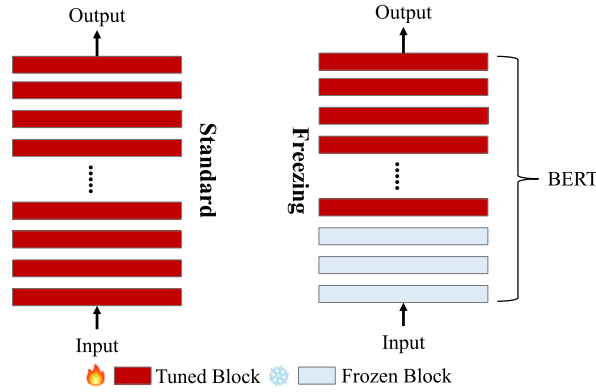
Specifically, we adopt the original BERT initialization, $\mathcal{N}(0, 0.02)$, to initialize the pooler layers and the top $\mathcal{L}_T \in \{1, \dots, 11\}$ BERT Transformer blocks, instead of relying on the pre-trained weights for all layers. During the fine-tuning process, we compare the BERT of Reinit-Layers with the BERT of standard layers, which are initialized by the pre-trained weights for all layers. The experiment results are shown in Table 1.

Observation results are summarized as follows. First, using Reinit-Layers can obviously improve the pre-trained BERT's generalization. In detail, during the fine-tuning process, the BERT of standard layers only achieves an accuracy of about 54% on the unseen BuzzFeedNews dataset, but the Reinit-Layers fine-tuning of BERT methods all achieve an accuracy of about 60% on

Table 1

Generalization performance on BuzzFeed using fine-tuned BERT with different initializations of the pre-trained BERT layers, which is trained on FakeNewsNet.

Method	Standard	Reinit $\mathcal{L}_T$ top layers										
		$\mathcal{L}_T = 1$	$\mathcal{L}_T = 2$	$\mathcal{L}_T = 3$	$\mathcal{L}_T = 4$	$\mathcal{L}_T = 5$	$\mathcal{L}_T = 6$	$\mathcal{L}_T = 7$	$\mathcal{L}_T = 8$	$\mathcal{L}_T = 9$	$\mathcal{L}_T = 10$	$\mathcal{L}_T = 11$
FakeNewsNet	100%	100%	100%	100%	100%	100%	100%	100%	99.71%	98.67%	98.02%	96.42%
BuzzFeedNews	54.39%	60.43%	64.84%	64.5%	65.93%	60.43%	59.89%	56.04%	53.19%	52.19%	52.06%	51.35%

**Fig. 3.** The Freezing strategy aims to evaluate the effect of freezing the parameters of the first few layers in the fine-tuning process, which means that the first few layers of the pre-trained BERT model are kept fixed and not updated during fine-tuning.**Table 2**

Generalization performance on BuzzFeed using fine-tuned BERT with different freezing of pre-trained BERT layers, which is trained on FakeNewsNet.

Method	Standard	Freezing $\mathcal{L}_B$ bottom layers										
		$\mathcal{L}_B = 1$	$\mathcal{L}_B = 2$	$\mathcal{L}_B = 3$	$\mathcal{L}_B = 4$	$\mathcal{L}_B = 5$	$\mathcal{L}_B = 6$	$\mathcal{L}_B = 7$	$\mathcal{L}_B = 8$	$\mathcal{L}_B = 9$	$\mathcal{L}_B = 10$	$\mathcal{L}_B = 11$
FakeNewsNet	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
BuzzFeedNews	54.39%	54.94%	53.74%	53.83%	53.83%	52.74%	52.19%	52.59%	52.04%	52.94%	52.89%	52.04%

the unseen BuzzFeedNews dataset. Second, although Reinit-Layers can improve the generalization performance, re-initializing more layers could also reduce the generalization performance. In detail, when $\mathcal{L}_T > 4$, the generalization performance decreases. The optimal value of $\mathcal{L}_T$ varies depending on the dataset.

5.1.2. Freezing of the pre-trained BERT layers

When the pre-trained BERT model is fine-tuned, it is common to freeze some layers of the BERT. Freezing refers to setting these layers as non-trainable, meaning the weights of the BERT are not updated during the fine-tuning. Freezing pre-trained layers could potentially enhance the model's efficacy. In detail, freezing prevents overfitting on the limited downstream task data by leveraging the general knowledge already captured. In the standard method of fine-tuning BERT, all layers are updated in fine-tuning, while in the freezing method, several lower layers of the pre-trained BERT are frozen. The weights of them are not updated during the fine-tuning.

Previous works (Nguyen, Le, Le, Pham, & Nguyen, 2023) have demonstrated that freezing of the pre-trained BERT layers can not only enhance the BERT's performance but also reduce the training cost of the BERT during the fine-tuning process. However, despite the widespread use of pre-trained models, there has been limited research on the specific effect of freezing certain layers on their overall generalization performance. Therefore, in this paper, we are interested in addressing this gap by conducting a thorough investigation of the impact of freezing BERT layers on the generalization ability of models for detecting fake news. To achieve this goal, as shown in Fig. 3, we design a comprehensive ablation study that explores the different freezing configurations and their corresponding effects on the models' accuracy and robustness. Through this analysis, we hope to provide valuable insights into the optimal freezing strategies for pre-trained models, which can help improve their generalization performance. Specifically, in the fine-tuning, we freeze the bottom $\mathcal{L}_B \in \{1, \dots, 11\}$ BERT Transformer blocks that are not updated. Thus, during the fine-tuning, we compare the Freezing-Layer fine-tuning BERT with the standard fine-tuning BERT, whose layers are updated. The experimental results are presented in Table 2.

Observation results are summarized as follows. First, using freezing layers even reduces the generalization of the model. In detail, the standard BERT achieves an accuracy of about 54% on the unseen BuzzFeedNews dataset, while the best of the Freezing-Layer method also only achieves an accuracy of about 54%. Secondly, the number of layers that a model freezes can significantly affect its ability to generalize in detecting fake news. Through our research, we have discovered that while freezing the pre-trained BERT layers may improve the training speed and efficiency of the model, it also has the potential to decrease its overall

Table 3

Generalization performance on BuzzFeed using fine-tuning BERT with different training iterations, which is trained on FakeNewsNet.

Method	Training Epoch-3	Training Epoch-6	Training Epoch-9	Training Epoch-12	Training Epoch-15	Training Epoch-18	Training Epoch-21
FakeNewsNet	100%	100%	100%	100%	100%	100%	100%
BuzzFeedNews	54.39%	56.04%	54.39%	53.79%	52.74%	51.94%	51.76%

generalization ability. Therefore, when designing models for fake news detection, achieving a balance between training efficiency and generalization performance is crucial.

5.1.3. Training iterations for fine-tuning

Howard and Ruder (2018) proposed a slanted triangular learning rate strategy, which is a widely adopted method for fine-tuning BERT models. During the strategy, the learning rate initially increases in some iterations and then declines. This approach has been found to be effective in fine-tuning BERT, allowing the BERT to adapt to the specific downstream task while avoiding overfitting. To apply the slanted triangular learning rate strategy, the training iterations must be predetermined. Lee, Cho, and Kang (2020) have indicated fine-tuning BERT using three epochs, which has become a standard practice in the field. However, recent research by Zhang et al. (2020) has indicated that extending the number of training epochs beyond three can improve the model performance. Despite these findings, the effect of training iterations on generalization has not been thoroughly explored. It is possible that increasing the number of iterations could lead to overfitting and reduce model generalization. It is necessary to explore the relationship between the training iterations and model generalization in the context of BERT fine-tuning to detect fake news. We study the effect of iterations on the generalization of the model, specifically in the context of fine-tuning BERT to detect fake news. Through our research, we have found that limiting the fine-tuning process to just three iterations can have a negative effect on model generalization. By conducting experiments with varying numbers of training iterations, we have determined that increasing the appropriate number of iterations significantly boosts the generalization capabilities. This is due to the fact that with each additional iteration, the model becomes more finely tuned to the specific task, and is better able to recognize and classify new and unseen data. Note that using more training iterations cannot effectively improve the generalization for fake news detection. Only using appropriate training iterations to fine-tune BERT can improve model generalization. Specifically, we adopt more training epochs to fine-tune BERT for detecting fake news. For the original FakeNewsNet dataset, three training epochs are equivalent to 491 training iterations. We expand the number of the training iterations to 491×2 , 491×3 , 491×4 , 491×5 , 491×6 , 491×7 , and 491×8 . In detail, we use 3 training epochs, 6 training epochs, 9 training epochs, 12 training epochs, 15 training epochs, 18 training epochs, and 21 training epochs to fine-tune the BERT on FakeNewsNet for fake news detection. Then we adopt their accuracy on another dataset, *i.e.*, BuzzFeedNews, as a metric to select the appropriate epochs. The Table 3 presents experimental results.

It is clear that using 6 training epochs can achieve the best generalization performance. And the generalization accuracy on an unseen dataset is used as the metric to verify that. Observation results are summarized as follows. First, using more training iterations can significantly boost the model generalization. In detail, when the training iterations are set to 491×2 , the fine-tuned BERT can achieve an accuracy of about 56%, which is higher than the fine-tuned BERT trained using 491 training iterations. Second, although using more training iterations can improve the generalization performance, but using more training iterations to fine-tune BERT reduces the generalization performance. Specifically, when the training iterations are set to 491×8 , the fine-tuned BERT can achieve an accuracy of about 52%, which is lower than the fine-tuned BERT trained using 491 training iterations. Only using appropriate training iteration can improve the generalization of the model. Specifically, a shorter training iteration will prevent the model from distinguishing fake news, while a longer training iteration will cause the model to overfit the training data, resulting in reduced generalization performance in detecting fake news. Hence, brutally increasing training epochs does lead to poor generalization performance. Hence, selecting the optimal number of training iterations for fine-tuning BERT is an important consideration that can obviously impact the generalization performance. Appropriately increasing the number of training iterations can boost the generalization performance of the BERT and achieve better experimental results on downstream tasks.

5.1.4. Baseline based on pre-trained BERT for fake news detection

According to the above-mentioned techniques, we have reached a conclusion on the baseline fine-tuning training setting on the FakeNewsNet dataset for detecting fake news by utilizing pre-trained BERT. To elaborate, in order to enhance the performance, we suggest implementing the original BERT initialization, $\mathcal{N}(0, 0.02)$, to initialize the pooler layers and the first top-4 BERT Transformer blocks. This initialization has been shown to be effective for pre-trained models and can help the model learn task-specific features more effectively. In terms of freezing the pre-trained BERT layers, we suggest freezing the first bottom-1 BERT Transformer blocks for the purpose of experimentation. The lower layers of pre-trained models like BERT tend to learn more general features. By leaving these layers frozen, we can allow the model to recognize patterns and features in the data that are broadly applicable across a wide range of tasks to improve the generalization performance. In terms of the training iterations used for fine-tuning, we propose utilizing a greater number of iterations to achieve optimal fine-tuning of BERT. Besides, initialization, freezing, and training iteration strategies can be combined to further improve model generalization. To verify their effectiveness, we evaluate generalization performance on BuzzFeedNews using BERT-based fake news detector under four combinations of these three strategies. Firstly, we combine these three strategies together. Under this combination, we first initialize the first 6 top layers of the pre-trained model. After that, we freeze the first 1 bottom layer of the model. Finally, the model is fine-tuned for 6 training iterations. Then, we continue

Table 4

Generalization performance on BuzzFeed using fine-tuning BERT with different combinations of initialization, freezing and, training iterations, which is trained on FakeNewsNet.

Reinit	Freezing	Iterations	BuzzFeedNews
✓	✓		66.4
✓		✓	67.5
	✓	✓	59.3
✓	✓	✓	68.1

to combine two of them respectively to evaluate the generalization performance of fine-tuning BERT for fake news detection. The experiment results are shown in Table 4.

It can be observed that three fine-tuning strategies are compatible and combining them could achieve better generalization. Employing these three strategies together results in the most effective generalization for BERT in detecting fake news. Overall, by incorporating these techniques, we can effectively improve the generalization of fine-tuned BERT for fake news detection and achieve better results on downstream tasks.

5.2. Adversarial fine-tuning training strategy for fake news detection

Adversarial training (Jia, Zhang, Wu, Wang & Cao, 2022) is a powerful method that is developed to enhance the model's robustness against adversarial examples. It involves injecting carefully crafted adversarial examples into the training data, thereby encouraging the model to learn more robust representations that could prevent adversarial perturbations effectively. Several studies (Jia et al., 2022; Wong, Rice, & Kolter, 2020) have demonstrated the effectiveness of adversarial training in enhancing model robust generalization.

In this paper, we study the impact of adversarial training on model generalization in fake news detection tasks. After conducting our research, our findings suggest that incorporating adversarial training could obviously boost model generalization. Building on this finding, we propose a novel adversarial fine-tuning training strategy based on feature regularization, called FRAT. In the proposed method, adversarial training principles are utilized to enhance the generalization of models in detecting fake news while incorporating feature regularization to further enhance the quality of the results.

Given the training dataset $D = \{(\mathbf{x}_i, y_i)_{i=1}^n\}$, which consists of the input data $\mathbf{x}_i$ and the corresponding label y_i , the classification model $f(\cdot, \theta)$ based on BERT, and the loss function $\mathcal{L}(f(\mathbf{x}_i, \theta), y_i)$, we formulate adversarial training for fake news detection as a minimax optimization problem. We can define the objective function as:

$$\begin{aligned} \min_{\theta} \mathbb{E}_{(\mathbf{x}_i, y_i) \sim D} [\max_{\delta} \mathcal{L}(f(\mathbf{x}_i + \delta, \theta), y_i)] \\ \text{s.t. } \|\delta\|_p \leq \epsilon, \end{aligned} \quad (1)$$

where δ represents the adversarial perturbation, and $\|\delta\|_p \leq \epsilon$ represents the perturbation boundary, which is used to control the strength of the perturbation. During the objective function, the inner maximization is used to generate the adversarial perturbations to maximize the loss function, while the outer minimization is used to optimize the parameters θ of the robust model to minimize the loss function. In this work, we adopt a common adversarial attack method, i.e. Fast Gradient Method (FGM), to conduct experiments. As presented in Fig. 4, it is clear that using FGM adversarial training techniques can significantly enhance the generalization for detecting fake news. Compared with the original fine-tuning BERT, the BERT combined with FGM adversarial training can achieve better generalization performance. In detail, on the unseen BuzzFeed, the original fine-tuning BERT only achieves the performance of about 54% accuracy. However, using FGM adversarial training to fine-tune BERT can achieve the performance of more than 60% accuracy when the epsilon strength is larger than 0.2. In particular, when epsilon strength is set to 0.8, FGM-AT achieves the highest generalization performance, which achieves a performance of about 68% accuracy. The experimental results indicate that the inclusion of adversarial training has a considerable impact on enhancing the overall performance of the BERT for detecting fake news, leading to improved model generalization.

To further boost the generalization, we propose a novel adversarial fine-tuning training strategy based on feature regularization. In detail, during the mini-max optimization problem, we incorporate an adversarial regularization loss at the feature level to regularize the parameters, thereby improving the generalization of the model. We can define the proposed objective function as:

$$\begin{aligned} \min_{\theta} \mathbb{E}_{(\mathbf{x}_i, y_i) \sim D} [\max_{\delta} [\mathcal{L}(f(\mathbf{x}_i + \delta, \theta), y_i) + \lambda \|f_F(\mathbf{x}_i + \delta, \theta) - f_F(\mathbf{x}_i, \theta)\|_2^2]] \\ \text{s.t. } \|\delta\|_p \leq \epsilon, \end{aligned} \quad (2)$$

where $f_F(\cdot, \theta)$ is the feature output of the model $f(\cdot, \theta)$, λ is the hyper-parameter. The proposed adversarial training strategy involves two steps: inner maximization and outer minimization. During the inner maximization step, the generated adversarial perturbation serves a dual purpose. It maximizes not only the original loss function, but also the L_2 distance between the adversarial examples and the corresponding clean samples at the model feature layer. This means that the perturbation modifies the input data in such a way that it is perceptually similar to the original input, while still inducing a high loss for the model. Similarly, during the outer minimization step, the model's parameters are adjusted to minimize the primary loss function and the L_2 distance between the adversarial examples and the corresponding clean samples at the model feature layer. This ensures that the model learns to resist

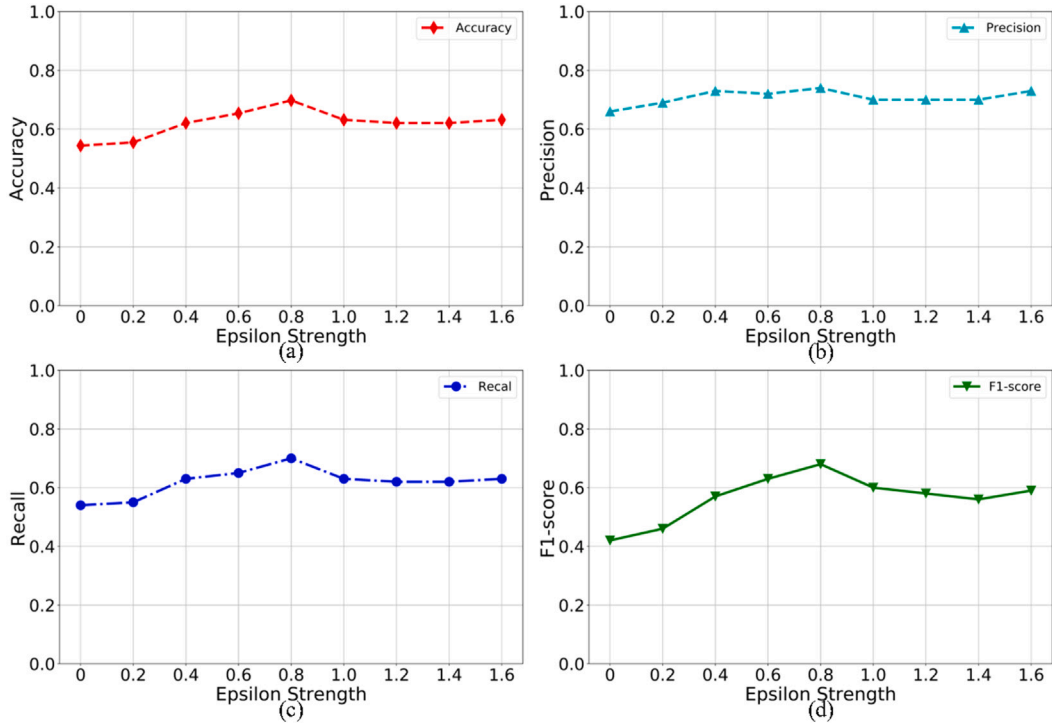


Fig. 4. Generalization performance on BuzzFeed of FGM adversarial training method for fake news detection with different attack strengths ($\epsilon_{\infty} = 0.0 \rightarrow 1.6$), which is trained on FakeNewsNet. The performance is evaluated by several evaluation metrics, including (a) Accuracy, (b) Precision, (c) Recall, and (d) F1-score.

Table 5

The details of the datasets used in the work.

Dataset	All size	Real size	Fake size	Source
FakeNewsNet	44,000	23,000	21,000	Twitter
BuzzFeedNews	2,282	1,669	613	Facebook
LIAR	12,791	7,133	5,658	Political
WELFake	72,134	35,028	37,106	Kaggle, BuzzFeed, Political, McIntire
KaggleFakeNews	16,600	8,213	8,387	Kaggle

adversarial attacks while still maintaining its ability to correctly classify clean samples. By adopting our adversarial training strategy, the loss landscape is smoothed out, which in turn enhances the model's generalization ability. Because the adversarial examples are more challenging input sets for the model to learn from, and thus the model becomes more robust to different types of input data, which can make the model able to better generalize to new and unseen data.

6. Experiments

A series of experiments are conducted to evaluate the effectiveness of the proposed method. They are fully detailed in this section. Specifically, we begin by introducing the specifics of several benchmark databases in Section 6.1, providing an overview of their key features and characteristics. We then move on to outline the experimental setup in Section 6.2, providing a clear and concise overview of our methodology and approach. An ablation study is presented in Section 6.3. We show the experimental results for detecting fake news in Section 6.5.

6.1. Benchmark databases

Following previous works (Sedik, Abohany, Sallam, Munasinghe, & Medhat, 2022; Zhang & Ghorbani, 2020), several fake news benchmark databases, which are widely used to detect fake news, are used to conduct experiments. As shown in Table 5, they consist of FakeNewsNet (Ahmed, Traore, & Saad, 2018), BuzzFeedNews, LIAR (Wang, 2017), WELFake (Verma, Agrawal, Amorim, & Prodan, 2021) and KaggleFakeNews.¹

¹ <https://www.kaggle.com/c/fake-news/data>

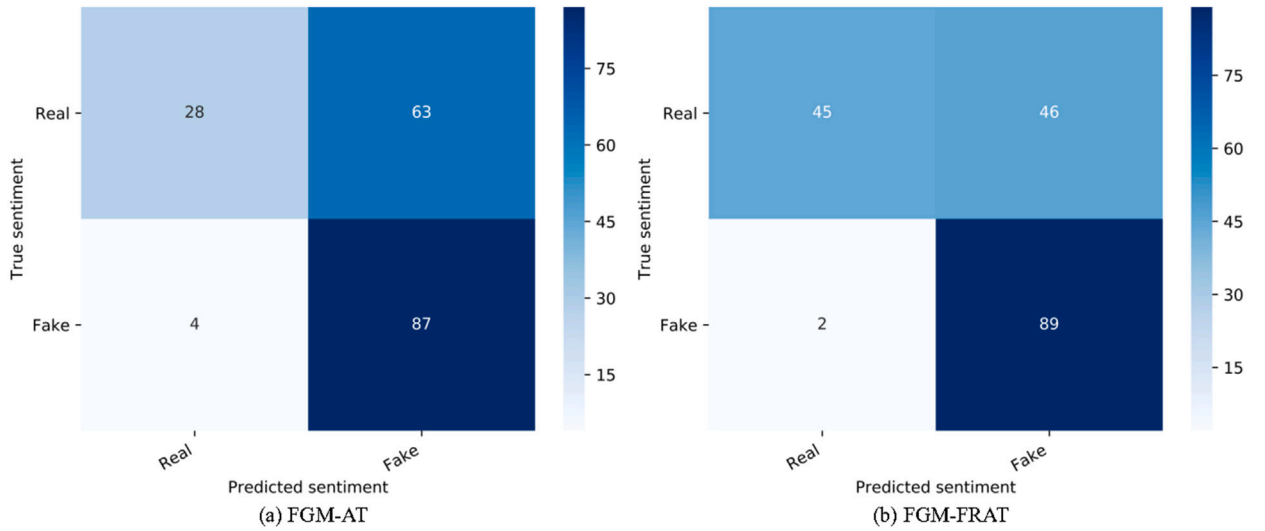


Fig. 5. Ablation study of the proposed method. The confusion matrix of generalization performance on BuzzFeed of BERT models that are trained on FakeNewsNet. (a) FGM-AT represents the adversarial training with FGM for fake news detection. (b) FGM-FRAT represents the adversarial training with FGM based on feature regularization for fake news detection.

The FakeNewsNet dataset is a dataset that includes fake and real political news reports, each with a link to the source and Twitter information. The Twitter information includes the person who shared the article and engagement metrics such as likes and retweets. It includes 44,000 real and fake news reports. The BuzzFeedNews dataset includes complete Facebook news, which is annotated by journalists. LIAR is collected from PolitiFact, which includes brief statements such as press releases, television, campaign speeches, and metadata. And the LIAR dataset is a well-known and difficult-to-detect fake news dataset, containing 12,791 short statements that have been manually labeled and obtained from POLITIFACT.COM. The WELFake dataset contains 37,106 fake and 35,028 real news articles, which is merged by four widely used news datasets: Reuters Kaggle, BuzzFeed, Political, and McIntire. The KaggleFakeNews dataset is obtained from Kaggle, which is a platform for data science competitions and the community. Specifically, the dataset of fake news, which we employed in this paper, includes around 16,600 rows of data, each representing an article from the internet. To evaluate the generalization of our FGM-FRAT, we adopt the FakeNewsNet database to fine-tune the BERT. Evaluation experiments are conducted on the LIAR, WELFake, and KaggleFakeNews databases.

6.2. Experimental setup

To adopt the BERT for detecting fake news, we utilized the pre-trained BERT from HuggingFace. This pre-trained model consists of 12 encoder layers. The hidden dimension of each layer is set to 768. The maximum token length is set to 512. We utilized these pre-trained text representations and fed them into a linear classification layer with a 0.1 dropout rate. We adopt cross-entropy loss to optimize the proposed method. The training batch size is set to 8. The initial learning rate is set to $3e-6$. And the training epoch is set to 3. We conduct all experiments on the NVIDIA RTX 3090 GPU. To evaluate the effectiveness of our FGM-FRAT, we utilized the commonly employed metrics of accuracy, F1-score, recall, and precision.

6.3. Ablation study

We first explore the impact of adversarial training on model generalization in detecting fake news and find that using adversarial training to detect fake news can improve the model generalization effectively. Specifically, we adopt FGM to generate adversarial examples for adversarial training, called FGM-AT. Based on FGM-AT, we propose a novel adversarial fine-tuning training strategy based on feature regularization, called FGM-FRAT. To explore the effects of the proposed feature regularization, we adopt pre-trained BERT to conduct an ablation study. In detail, we adopt FGM-AT with the proposed feature regularization to fine-tune BERT on FakeNewsNet and evaluate their generalization on BuzzFeed. The results of the confusion matrix are presented in Fig. 5. It is obvious that the proposed FGM-FRAT method has demonstrated superior performance in terms of generalization when compared to the FGM-AT method for detecting fake news. Specifically, when evaluated on the BuzzFeed dataset, which is an unseen dataset, as for the metrics of accuracy, the FGM-AT method only achieves an accuracy rate of about 64%. The proposed FGM-FRAT method obtains an accuracy rate of approximately 74%, which represents a significant improvement over the FGM-AT method. Experimental evaluations provide strong evidence that incorporating the proposed feature regularization approach can lead to a

Table 6

Generalization performance, which includes metrics of accuracy, precision, recall, and F1-score, on a series of unseen databases, which consist of BuzzFeed, LIAR, WELFake, and KaggleFakeNews using pre-trained BERT models with different fine-tuning strategies, which are trained on FakeNewsNet.

Method	BuzzFeedNews	LIAR	WELFake	KaggleFakeNews
BERT	54%/76%/54%/42%	58%/49%/49%/48%	82%/86%/82%/82%	51%/48%/50%/39%
BERT-Reinit	66%/78%/66%/62%	73%/65%/73%/67%	83%/85%/83%/83%	49%/46%/49%/42%
BERT-Freezing	55%/69%/55%/45%	73%/65%/73%/68%	82%/84%/82%/82%	51%/47%/51%/40%
BERT-Iterations	56%/73%/56%/46%	70%/64%/70%/66%	84%/85%/83%/83%	49%/46%/49%/43%
BERT-FGM-AT	63%/73%/63%/59%	69%/64%/69%/66%	83%/85%/82%/82%	49%/49%/49%/49%
BERT-FGM-FRAT (Ours)	74%/81%/72%/74%	75%/65%/75%/67%	84%/85%/84%/84%	61%/68%/61%/64%

considerable enhancement in the model's generalization ability for detecting fake news, particularly when fine-tuning BERT. Overall, the proposed FGM-FRAT can boost the performance to detect fake news and contribute to the development of more accurate and reliable approaches in this domain.

6.4. Comparisons with other fine-tuning strategies

In this paper, we propose a novel adversarial fine-tuning strategy based on feature regularization to improve generalization. We compare the proposed FGM-FRAT with several fine-tuning strategies to evaluate its effectiveness. They include initialization of the BERT layers, freezing of the BERT layers, training iterations, and FGM-AT. In detail, we adopt the basic database, FakeNewsNet, to fine-tune the BERT and evaluate the generalization performance on a series of unseen databases, which consist of BuzzFeed, LIAR, WELFake, and KaggleFakeNews. We use the training settings from Section 5, which achieve the best generalization performance, to perform comparative experiments. Specifically, for the initialization of the pre-trained BERT layers, the top $\mathcal{L}_T$ layers are set to 4, which are initialized by the original BERT initialization instead of the weights of pre-trained BERT. For the freezing of the pre-trained BERT layers, the bottom $\mathcal{L}_B$ layers are set to 1, which is not updated during the fine-tuning process. For the training iterations for fine-tuning, the number of the training iterations is set to 491×3 , i.e., the training epoch is set to 6. Table 6 presents the experimental results.

Upon careful observation, it is evident that the proposed fine-tuning strategy surpasses other existing strategies in terms of generalization performance, thus showcasing the ability of pre-trained BERT combined with the proposed FRAT to significantly enhance model generalization for fake news detection. As for the metrics of accuracy, the original BERT achieves the detection accuracy of 100% on FakeNewsNet, but it only achieves an accuracy of approximately 54%, 58%, 82%, and 51% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. Using the existing fine-tuning strategies can significantly boost the generalization of the BERT-based fake news detectors. For example, the best-performing fine-tuning strategy among previous approaches could only achieve an accuracy of approximately 66%, 73%, 83%, and 49% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively, which improves the average accuracy by about 6% on these unseen datasets. But, the proposed method can obtain accuracy of about 74%, 75%, 84%, and 61% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively, which improves the average accuracy by about 12% on these unseen datasets (61% VS 73%). These results indicate the proposed method can significantly enhance the generalization of the BERT-based fake news detectors.

As for the metrics of precision, the original BERT only achieves an accuracy of approximately 76%, 49%, 86%, and 48% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. And the BERT-Reinit achieves an accuracy of approximately 78%, 65%, 85%, and 46% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. But our proposed BERT-FGSM-FRAT achieves an accuracy of approximately 81%, 65%, 85%, and 68% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively, which achieves the best performance. Particularly, on the KaggleFakeNews dataset, the previous methods only achieve precision accuracy below 50%, but our method can achieve a precision accuracy of 68%. As for the metrics of recall, the original BERT only achieves an accuracy of approximately 54%, 49%, 82%, and 50% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. And BERT-Freezing achieves an accuracy of approximately 55%, 73%, 82%, and 51% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. But our proposed BERT-FGSM-FRAT achieves an accuracy of approximately 72%, 75%, 84%, and 61% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. As for the metrics of F1-score, the original BERT only achieves an accuracy of approximately 42%, 48%, 82%, and 39% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. And the BERT-Iterations achieves an accuracy of approximately 46%, 66%, 83%, and 43% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. But our proposed BERT-FGSM-FRAT achieves an accuracy of approximately 74%, 67%, 84%, and 64% on BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, respectively. These experimental results indicate that the proposed FGSM-FRAT achieves superior performance in other evaluation metrics.

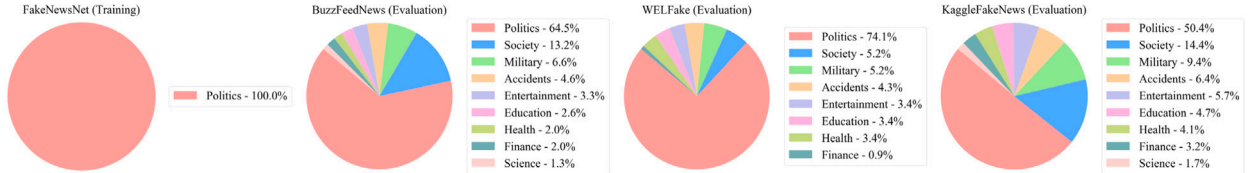
6.5. Comparisons with state-of-the-art fake news detection methods

To further evaluate the effectiveness of the proposed BERT-FGM-FRAT, we compare our BERT-FGM-FRAT with a series of state-of-the-art Fake News detection methods, including CNN (Shrivastava et al., 2020), GRUs (Li, Guo, Wang, & Zheng, 2021), LSTM (Van Houdt, Mosquera, & Nápoles, 2020), BERT (Wani, Joshi, Khandve, Wagh, & Joshi, 2021), RoBERTa-MLP (Shifath, Khan, & Islam, 2021), and RoBERTa-CNN (Samadi, Mousavian, & Momtazi, 2021). In detail, we adopt the FakeNewsNet database to

Table 7

Generalization performance, including metrics of accuracy, precision, recall, and F1-score, on a series of unseen databases consisting of BuzzFeed, WELFake, and KaggleFakeNews using pre-trained BERT models with state-of-the-art fake news detection methods trained on FakeNewsNet.

Method	BuzzFeedNews	WELFake	KaggleFakeNews	Average
CNN (Shrivastava et al., 2020)	58%/71%/59%/52%	81%/85%/82%/81%	52%/53%/51%/39%	64%/70%/64%/57%
GRUs (Li et al., 2021)	51%/59%/52%/39%	82%/85%/83%/83%	51%/50%/50%/48%	61%/65%/62%/57%
LSTM (Van Houdt et al., 2020)	54%/76%/54%/42%	82%/85%/82%/82%	51%/45%/49%/37%	62%/69%/62%/54%
BERT-MLP (Wani et al., 2021)	54%/76%/54%/42%	82%/85%/82%/82%	51%/48%/50%/39%	62%/70%/62%/54%
RoBERTa-MLP (Shifath et al., 2021)	64%/79%/64%/58%	82%/85%/82%/82%	51%/49%/50%/47%	66%/71%/65%/62%
RoBERTa-CNN (Samadi et al., 2021)	65%/78%/65%/61%	82%/85%/83%/83%	51%/49%/49%/48%	66%/71%/66%/64%
BERT-FGM-FRAT (Ours)	74%/81%/72%/74%	84%/85%/84%/84%	61%/68%/61%/64%	73%/78%/72%/74%

**Fig. 6.** Proportion of fake news in different domains on the BuzzFeedNews, WELFake, and KaggleFakeNews datasets.**Table 8**

The MMD distance between the three target datasets, including BuzzFeedNews, WELFake, and KaggleFakeNews and the source dataset, including FakeNewsNet, separately.

Dataset	BuzzFeedNews	WELFake	KaggleFakeNews
MMD	0.1612	0.0252	0.2327

train these fake news detection models. Then the generalization performance of the well-trained models is evaluated on a series of unseen databases, which include BuzzFeed, WELFake, and KaggleFakeNews. The metrics of accuracy, precision, recall and F1-score are used to evaluate the generalization performance. The experimental results are shown in Table 7. It is clear that compared with previous fake news detection methods, the proposed method achieves the best generalization performance across all datasets. Specifically, the classic fake news method, *i.e.*, LSTM (Van Houdt et al., 2020), obtains the accuracy of 54%, 82%, and 51% on the BuzzFeedNews, WELFake and KaggleFakeNews, respectively. However, the proposed BERT-FGM-FRAT can achieve accuracy of 74%, 84%, and 61% on the BuzzFeedNews, WELFake and KaggleFakeNews, respectively. Even compared to the state-of-the-art method, *i.e.*, RoBERTa-CNN (Samadi et al., 2021), the proposed BERT-FGM-FRAT improves the accuracy by 9%, 2% and 10% on the BuzzFeedNews, WELFake and KaggleFakeNews, respectively. Hence, previous fake news detection methods only achieved an average accuracy of 65% on unseen datasets, but the proposed BERT-FGM-FRAT can achieve an average accuracy of 73% on unseen datasets. It indicates that the proposed method can significantly improve the generalization performance of the fake news detection models. In terms of other metrics, we observe similar results on the metric of accuracy.

Note that the reason behind the different performance on different datasets is mainly as follows. The FakeNewsNet dataset used for training only consists of fake and real political news. But the BuzzFeedNews, WELFake, and KaggleFakeNews datasets, which are used to evaluate generalization, include multi-domain fake and real news. The results are shown in Fig. 6.

It can be observed that political news accounts for 64.5% of the entire data in the BuzzFeedNews dataset, 74.1% of the entire data in the WELFake dataset, and 50.4% of the entire data in the KaggleFakeNews dataset. Previous works (Liang et al., 2022; Nordberg et al., 2020) have indicated that each domain has different contents and styles, making it difficult for a model trained in one domain to generalize to another domain. Hence, different proportions of political news in different datasets lead to different generalization of the model on different datasets. Moreover, we adopt Maximum Mean Discrepancy (MMD) (Gretton, Borgwardt, Rasch, Schölkopf, & Smola, 2012) to measure the distribution distance between the three target datasets, including BuzzFeedNews, WELFake, and KaggleFakeNews, and the source dataset, including FakeNewsNet, separately. The results are shown in Table 8.

Also, it can be observed that the distribution distance between the FakeNewsNet dataset and the WELFake dataset is the smallest, while the distribution distance between the FakeNewsNet dataset and the KaggleFakeNews dataset is the largest. It indicates the reason why the model performs best on the WELFake dataset and performs worst on the KaggleFakeNews dataset. In this paper, we formulate adversarial training for fake news detection as a minimax optimization problem. The inner maximization is used to generate the adversarial perturbations to maximize the loss function, while the outer minimization is used to optimize the parameters of fake news detection model to minimize the loss function. Previous works (Guo, Wang, Qi, & Shi, 2023; Sun, Advani, Spruston, Saxe, & Fitzgerald, 2023) have indicated that the model exhibits poor generalization due to its overfitting to the source dataset. To explore the impact of our proposed use of adversarial examples on overfitting, we visualize the changes in loss during our training

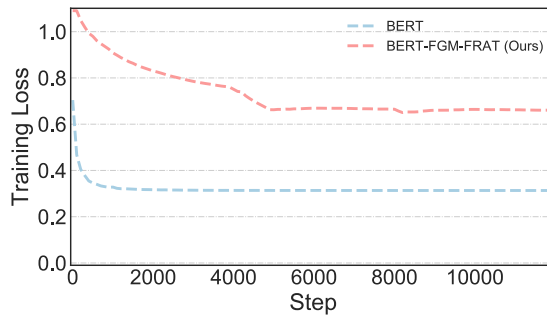


Fig. 7. Visualization of the evolution of loss during training of fake news detection model.

Table 9

Generalization performance, which includes metrics of accuracy, precision, recall, and F1-score, on an unseen BuzzFeed dataset using pre-trained RoBERTa-MLP trained on FakeNewsNet.

Dataset	Method	Accuracy	Precision	Recall	F1-score
BuzzFeed	RoBERTa-MLP	64%	79%	64%	58%
	RoBERTa-MLP-FGM-FRAT (Ours)	77%	79%	77%	72%

Table 10

Generalization performance, which includes metrics of accuracy, precision, recall, and F1-score, on an unseen BuzzFeed dataset, using BERT models with advanced multi-domain fake news detection methods trained on KaggleFakeNews.

Dataset	Method	Accuracy	Precision	Recall	F1-score
BuzzFeed	BERT	64%	44%	65%	52%
	MDFEND (Nan et al. 2021)	69%	51%	69%	59%
	FuFEND (Liang et al. 2021)	70%	58%	70%	60%
	BERT-FGM-FRAT (Ours)	75%	68%	75%	71%

process. The results are shown in Fig. 7. It is shown that using adversarial examples can alleviate the overfitting of the fake news detection model on the source dataset.

6.6. Extension of the proposed method

To explore the extension of the proposed method, we also adopt the proposed FGM-FRAT to fine-tune another pre-trained language model, i.e., RoBERTa-MLP to detect fake news. In detail, we adopt the FakeNewsNet database to conduct comparative experiments. Then the generalization performance of the well-trained models is evaluated on the unseen BuzzFeed dataset. The metrics of accuracy, precision, recall, and F1-score are used to evaluate the generalization performance. The experimental results are shown in Table 9.

It is clear that the proposed RoBERT-MLP-FGM-FRAT performs better than the original RoBERT-MLP on the unseen BuzzFeed, which indicates the proposed FGM-FRAT can greatly improve the generalization of the RoBERT-MLP. In detail, the original RoBERT-MLP only achieves an accuracy of 64%, precision of 79%, recall of 64%, and F1-score of 58%. While the RoBERT-MLP combined with the proposed FGM-FRAT achieves an accuracy of 77%, precision of 79%, recall of 77%, and F1-score of 72%. The experimental results show that the proposed method could be effectively extended to other pre-trained language models.

Besides, we also compare the proposed method with some advanced multi-domain fake news detection methods, which include MDFEND (Nan et al., 2021) and FuFEND (Liang et al., 2022), to evaluate the effectiveness of the proposed method. In detail, we adopt the multi-domain KaggleFakeNews dataset to train the BERT and the multi-domain BuzzFeed to evaluate the generalization. Note that previous experiments are conducted on the single-domain FakeNewsNet dataset, which only consists of political news. Since the original KaggleFakeNews and BuzzFeed datasets do not contain multi-domain label information, following the works (Liang et al., 2022; Nan et al., 2021), we divide the news of KaggleFakeNews and BuzzFeed into 9 domains. Then we perform comparative experiments on the new KaggleFakeNews and BuzzFeed datasets. The results are shown in Table 10. It demonstrates that using multi-domain fake news detection methods can improve generalization, but compared with advanced MDFEND and FuFEND, the proposed method achieves better generalization performance. Specifically, the original RoBERT-MLP only achieves an accuracy of 64%, precision of 44%, recall of 65%, and F1-score of 52%. MDFEND (Nan et al., 2021) achieves an accuracy of 69%, precision of 51%, recall of 69%, and F1-score of 59%. FuFEND (Liang et al., 2022) achieves an accuracy of 70%, precision of 58%, recall of 70%, and F1-score of 60%. While the proposed method achieves an accuracy of 75%, precision of 68%, recall of 75%, and F1-score of 71%. The experimental results indicate that the proposed method has more advantages than multi-domain fake news detection methods in improving model generalization.

Besides, to more comprehensively evaluate the performance of the proposed method, we expanded our research to cover broader Chinese fake news collections, including WeChat dataset (Wang et al., 2020) and Weibo21 dataset (Nan et al., 2021), for our

Table 11

Generalization performance, which includes metrics of accuracy, precision, recall, and F1-score, on broader Chinese fake news datasets.

Method	WeChat Dataset				Weibo21 Dataset			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
BERT-MLP (Wani et al., 2021)	71%	63%	71%	66%	66%	43%	66%	52%
RoBERTa-MLP (Shifath et al., 2021)	76%	73%	76%	74%	67%	45%	67%	54%
RoBERTa-CNN (Samadi et al., 2021)	77%	74%	77%	76%	68%	46%	68%	55%
BERT-FGM-FRAT (Ours)	85%	74%	85%	79%	79%	63%	79%	70%

Table 12

Comparison results with more leading models in fake news detection, which include metrics of accuracy, precision, recall, and F1-score, on an unseen BuzzFeed dataset using pre-trained RoBERTa-MLP, which is trained on FakeNewsNet.

Dataset	Method	Accuracy	Precision	Recall	F1-score
BuzzFeed	RoBERTa-MLP	64%	79%	64%	58%
	RoBERTa-MLP-FGM-FRAT (Ours)	77%	79%	77%	72%
	ALBERT	62%	65%	62%	60%
	ALBERT-FGM-FRAT (Ours)	74%	76%	74%	73%
	XLNET	64%	72%	64%	58%
	XLNET-FGM-FRAT (Ours)	75%	76%	75%	75%

Table 13

Generalization performance on the task of movie review sentiment analysis, which includes metrics of accuracy, on a series of unseen MR using pre-trained BERT trained on IMDb.

Method	IMDb	MR
BERT	92.0%	60.2%
BERT-FGM-FRAT (Ours)	91.7%	67.7%

experimental analysis. The experimental results are shown in Table 11. It can be observed that the proposed method still outperforms the previous fake news detection methods on non-English fake news datasets. Specifically, the accuracy of the proposed BERT-FGM-FRAT is increased by 19.7% on the WeChat dataset and by 19.6% on the Weibo21 dataset. It is estimated that the proposed method has more advantages than previous fake news detection methods in improving the model generalization on non-English fake news datasets. Then we adopt a series of leading models, which include RoBERTa (Samadi & Momtazi, 2023), ALBERT (Capuano, Fenza, Loia, & Nota, 2023), and XLNET (Aljrees et al., 2023), to conduct comparison experiments. The results are shown in Table 12. It can be observed that the proposed method can significantly help advanced models improve generalization performance in detecting fake news. Specifically, the proposed RoBERTa-MLP-FGM-FRAT model demonstrates a significant improvement in accuracy, recording a 20.3% increase compared to the baseline RoBERTa-MLP model. Similarly, when compared to the baseline ALBERT model, the accuracy of the ALBERT-FGM-FRAT model is enhanced by 19.3%. Additionally, the XLNET-FGM-FRAT model outperforms the baseline XLNET model, with an accuracy increase of 17.2%. It can be found that the proposed method can significantly help advanced models improve the generalization performance in detecting fake news.

Moreover, we also explore the feasibility of the proposed method on other text classification tasks, such as movie review sentiment analysis. In detail, we adopt a classic dataset for movie review sentiment classification, *i.e.*, Internet Movie Database (IMDb) (Maas et al., 2011), to fine-tune the original BERT and the proposed BERT-FGM-FRAT. Then we use another dataset *i.e.*, Movie Review (MR) (Pang, Lee, & Vaithyanathan, 2002), to evaluate the generalization performance of these well-trained models. The IMDb dataset consists of 25,000 negative reviews and 25,000 positive reviews. We adopt half of them to train the sentiment classification models and the remaining half to evaluate the performance. The MR dataset includes 10,662 movie reviews, which are used to evaluate the generalization performance of the well-trained sentiment classification models. The results are shown in Table 13. It is clear that compared with the original BERT, the proposed BERT-FGM-FRAT achieves comparable accuracy performance on the IMDb and better generalization performance on the unseen MR. In detail, BERT obtains an accuracy of 92.0% on the IMDb and an accuracy of 60.2% on the MR. While our BERT-FGM-FRAT obtains an accuracy of 91.7% on the IMDb and an accuracy of 67.7% on the MR. It indicates that the proposed method also could be used to improve model generalization on other text classification tasks.

7. Research implications

The work on the fine-tuning of BERT models for detecting fake news has several important implications: (1) Addressing overfitting issues for fake news detection: By exploring various fine-tuning strategies, such as the initialization, freezing, and the number of fine-tuning iterations, the study contributes to the development of more robust models that perform well on unseen data for fake news detection. The initialization improves model generalization by removing the top-layer parameters of the pre-trained model that are specific to the pre-training task. Specifically, firstly, it reduces the risk of overfitting to the specific features or noise present in the pre-training task that may not be relevant to the new task. Secondly, it encourages the model to rely more on the generalized feature representations learned in the lower and middle layers, which are more adaptable to different tasks. The freezing underlying

parameters of a pre-trained model enhances its generalization capabilities because it preserves the generalized feature extraction layers developed during pre-training. These layers have learned to recognize patterns and features in the data that are broadly applicable across a wide range of tasks. By freezing these parameters, the model maintains its ability to capture these common features, which are crucial for performing well on new and unseen data. Short training iterations hinder the model's ability to identify fake news, whereas longer iterations result in overfitting the training data, reducing the model's generalization capability in detecting fake news. Hence, the appropriate number of training iterations can enhance the model's effectiveness in generalization. The ablation experiments within a certain range in this paper have found the appropriate number of training iterations, which improved the model's generalization capability in detecting fake news. (2) Improved generalization accuracy in fake news detection: By adopting the proposed adversarial fine-tuning strategy (FGM-FRAT), the study significantly enhances the generalization ability of the fine-tuning BERT model in detecting fake news. Experiments on various benchmark databases show that the generalization accuracy of BERT has improved from 61% to 73% using the proposed FGM-FRAT method. (3) Broader applicability: The study indicates that the proposed adversarial fine-tuning strategy can be extended to other pre-trained language models and to a variety of text classification tasks, suggesting a wide range of potential applications beyond fake news detection. In summary, this research has significant implications for enhancing the generalization of models used for detecting fake news, contributing to broader efforts in maintaining the accuracy and reliability of information in the digital age.

8. Conclusion

In this work, we study the efficacy of several different fine-tuning strategies for enhancing the generalization of fine-tuned BERT in the context of detecting fake news. Our experiments show that the combination of initialization and freezing of model parameters along with an appropriate number of training iterations yields the best baseline training setting for fine-tuned BERT. Moreover, we investigate the impact of adversarial training on model generalization and propose a novel adversarial fine-tuning approach known as FRAT, which utilizes feature regularization. Through extensive experiments on benchmark databases, which include FakeNewsNet, BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, we show that our proposed fine-tuning strategy performs better than existing methods for detecting fake news. The proposed method can outperform state-of-the-art fake news detection methods. In conclusion, our findings emphasize the importance of employing a suitable combination of techniques and utilizing adversarial training with feature regularization for improving the generalization performance of BERT in fake news detection. Moreover, the proposed method can be used to improve model generalization of other pre-trained language models. The proposed method also can be used to boost model generalization on other text classification tasks.

9. Ethical statement

This study aims to boost generalization of fine-tuning BERT for fake news detection, while adhering to ethical principles. In detail, we have adhered to the highest ethical standards in the study of fake news detection using fine-tuned BERT models. We recognize the sensitivity and potential societal impact of our study, particularly in the context of fake news and its detection. We have ensured that our experiments, methods, and data processing practices respect privacy, consent, and data protection norms, especially when dealing with real-world datasets. Our research employs publicly available datasets, such as FakeNewsNet, BuzzFeedNews, LIAR, WELFake, and KaggleFakeNews, for the purpose of improving the accuracy and generalization of fake news detection models. We have carefully considered the implications of our work, ensuring that it does not inadvertently promote or enable malicious use, such as the suppression of free speech or the targeting of legitimate news sources under the guise of fake news detection. The novel adversarial fine-tuning approach, FRAT, that we propose is designed with the intention of enhancing the reliability and fairness of fake news detection systems, minimizing biases that may arise from training data or model assumptions. We advocate for the transparent, responsible, and ethical use of our findings, emphasizing that our proposed method should be applied in a manner that supports the integrity of information and contributes positively to societal discourse. We also acknowledge the broader implications of our work on model generalization across other pre-trained language models and text classification tasks. We encourage future research in this area to adopt similar ethical considerations, particularly in the development and application of advanced AI techniques in sensitive domains. In summary, our commitment to ethical research practices underpins our approach to studying and improving fake news detection with BERT and other pre-trained language models. We aim to contribute to the field in a way that is respectful, responsible, and beneficial to the society.

CRedit authorship contribution statement

Simeng Qin: Writing – review & editing, Writing – original draft, Visualization, Software, Data curation. **Mingli Zhang:** Supervision, Funding acquisition, Formal analysis, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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